AP PRECALCULUS

Unit 1 • Worksheet 1.1A • Answer key

Worksheet 1.1A • Change in Tandem

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The table gives values of f. Which statement is consistent with the table?

x	0	2	4	6	8
$f(x)$	3	8	11	12	12.5

(A) f is increasing and concave up.

(B) f is increasing and concave down.

(C) f is decreasing and concave up.

(D) f is decreasing and concave down.

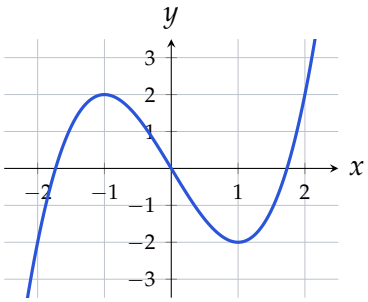
Solution

Answer: (B). Outputs increase; the changes $5, 3, 1, 0.5$ decrease, so the rate of change is decreasing: concave down.

Question 2

NO CALCULATOR

The graph of g is shown. On which interval is g decreasing and concave up?



(A) $(-1, 0)$
(B) $(0, 1)$
(C) $(1, 2)$
(D) $(-2, -1)$

Solution

Answer: (B). $g(x) = x^3 - 3x$ decreases on $(-1, 1)$. It is concave up to the right of $x = 0$. Both hold on $(0, 1)$.

Question 3

NO CALCULATOR

Which statement must be true if the rate of change of h is negative and increasing on an interval?

(A) h is increasing and concave down.
(B) h is decreasing and concave down.
(C) h is decreasing and concave up.
(D) h is increasing and concave up.

Solution

Answer: (C). Negative rate of change: decreasing. Increasing rate of change: concave up.

Question 4

NO CALCULATOR

What are the zeros of $p(x) = x(x-3)(2x+5)$?

(A) $0,\ 3,\ 5$
(B) $0,\ -3,\ \frac{5}{2}$
(C) $3,\ -\frac{5}{2}$
(D) $0,\ 3,\ -\frac{5}{2}$

Solution

Answer: (D). Set each factor equal to 0: $x = 0$, $x = 3$, $2x + 5 = 0 \Rightarrow x = -\frac{5}{2}$.

Question 5 (Free response) NO CALCULATOR

A tank is filling with water. The volume $V(t)$, in liters, is recorded every 10 minutes.

t (min)	0	10	20	30	40
$V(t)$ (L)	40	100	150	190	220

(a) Is V increasing or decreasing on $[0, 40]$? Explain.
(b) Find the change in volume over each 10-minute interval. Is the graph of V concave up or concave down? Explain.
(c) What does your answer to (b) say about how fast the tank is filling?

Solution
(a) Increasing: as t increases, $V(t)$ increases.
(b) $60, 50, 40, 30$. The changes over equal intervals decrease, so the rate of change is decreasing: concave down.
(c) The tank is filling more and more slowly.

Question 6 (Free response) NO CALCULATOR

Sketch a graph of a function that is increasing on $(-\infty, 2)$, decreasing on $(2, \infty)$, concave down everywhere, and has zeros at $x = -1$ and $x = 5$.

Solution
Any graph shaped like an upside-down parabola with its highest point at $x = 2$, crossing the x-axis at -1 and 5; for example $f(x) = -(x+1)(x-5)$, whose vertex is $(2, 9)$.